

PHYSICS 550
Homework #3
Due on Sept 22, 2023

1. G 2.35 (10 points)

2. G 2.47 (10 points)

3. (10 points)

An experimenter asks for funds to observe visually through a microscope the quantum behavior of a small oscillator. According to his proposal, the oscillator consists of an object 10^{-4}cm in diameter with an estimated mass of 10^{-12}g . It vibrates on the end of a thin fiber with a maximum amplitude of 10^{-3}cm and frequency 10^3Hz . You are referee for the proposal.

- (a) What is the approximate quantum number for the system in the state described?
- (b) What would be its energy in eV if it were in its lowest-energy state? Compare with the average thermal energy (25meV) of air molecules at room temperature.
- (c) What would be its classical amplitude of vibration if it were in its lowest-energy state? Compare this with the wavelength of visible light (500 nm) by which it is presumably observed.
- (d) Would you, as referee of this proposal, recommend award of a grant to carry out this research?